

ME 161 BASIC MECHANICS

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**COLLEGE OF ENGINEERING
KNUST-Kumasi**

Instruction Format

-  Lectures (2 hrs)
-  Tutorials (1 hr)

Assessment

- Assignments (4)
- Quizzes
- Attendance
- Mid Semester Exams (?)
- End of Semester Exams

Class Regulations

- Be punctual
- No use of mobile phones in class

References

- Hibbeler, R.C., *Engineering Mechanics: Statics and Dynamics*
- Beer et al., *Vector Mechanics for Engineers*

Outline of Module 1

□ Introduction

- What is Mechanics?
- Fundamental Concepts & Principles
- Systems of Units
- Numerical Calculations
- Problems

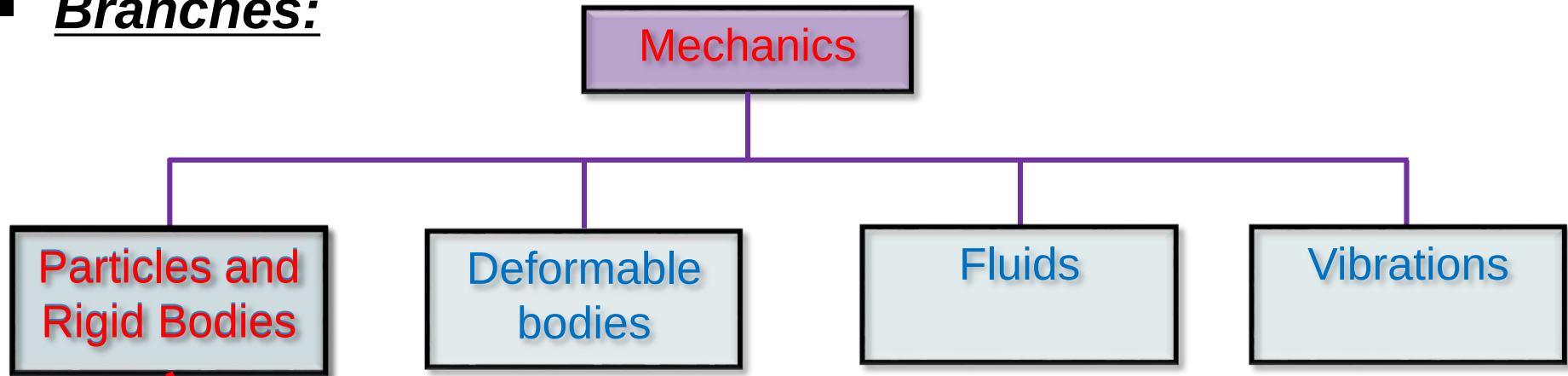
□ Vectors (Part 1)

- Force Vectors
- Addition of Force Vectors

What is Mechanics?

- **Mechanics** is a branch of the physical sciences that is concerned with the state of rest or motion of bodies that are subjected to the action of forces.

- **Branches:**



- **Statics:** deals with the equilibrium of bodies. That is, those that are either at rest or move with a constant velocity.
- **Dynamics:** is concerned with the accelerated motion of bodies.

Fundamental Concepts & Principles

A. Physical Quantities

- Four fundamental/basic physical quantities (or dimensions):
 - Length
 - Time
 - Mass
 - Force – It represents the action, pull or push, exerted by one body on another. A force is a **vector quantity**.

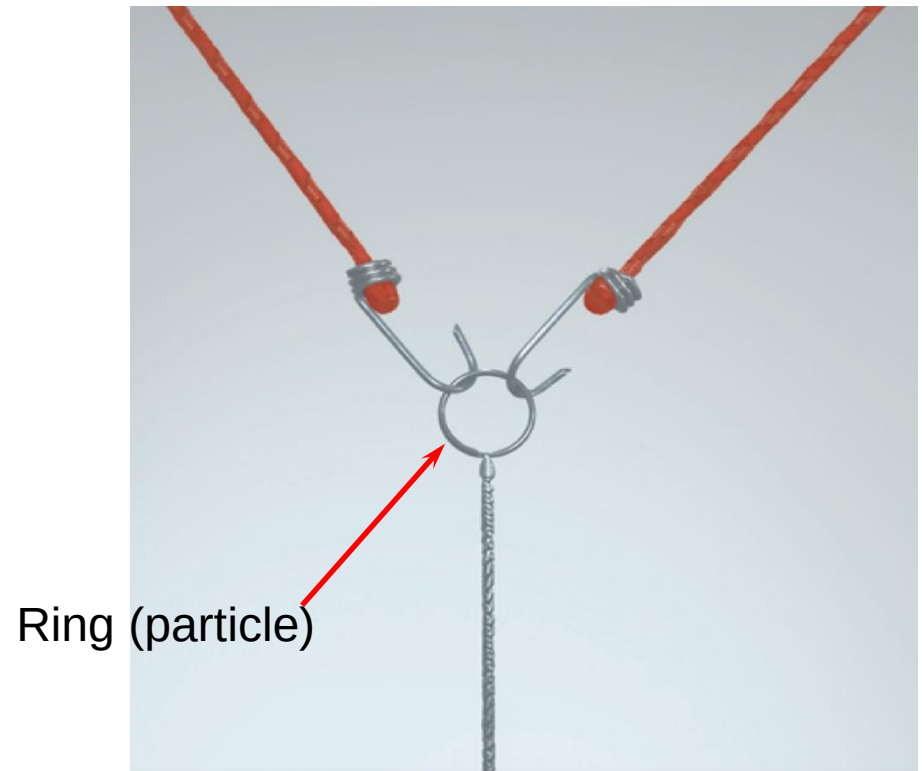
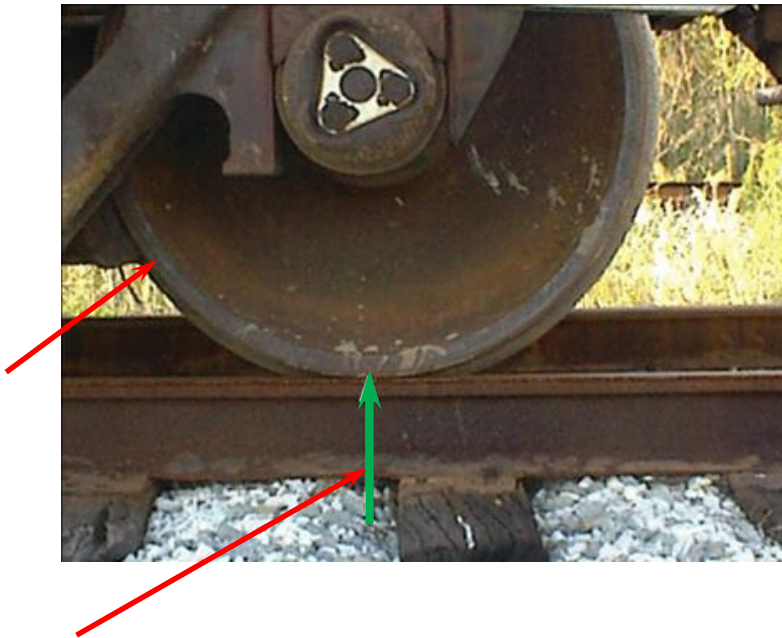
The four basic quantities are related by Newton's 2nd Law, $\mathbf{F} = m\mathbf{a}$.

- **Vector quantity** has magnitude and direction (e.g. velocity).
- **Scalar quantity** has only magnitude (e.g. speed).

Fundamental Concepts & Principles

B. Idealization/Models

- Particle – It has a mass, but with negligible size.
- Rigid Body – It is a solid body shape that does not deform.
- Concentrated Force/Load
- Distributed Force/Load

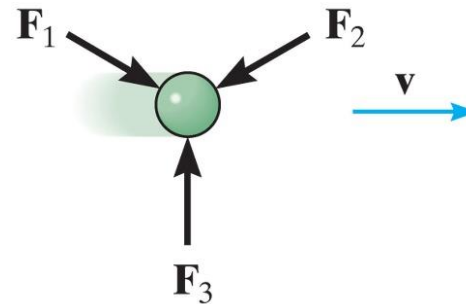


Fundamental Concepts & Principles

C. Newton's Three Laws of Motion

- **First Law** – If the resultant force on a particle is zero, the particle will remain at rest or continue to move in a straight line with a constant velocity.

$$\sum \mathbf{F} = \mathbf{0} \quad \text{OR} \quad \mathbf{F}_R = \mathbf{0}$$

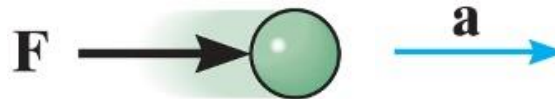


Equilibrium



- **Second Law** – A particle acted upon by unbalanced force experiences an acceleration that has the same direction as the force and a magnitude that is directly proportional to the force.

$$\mathbf{F} = m\mathbf{a}$$



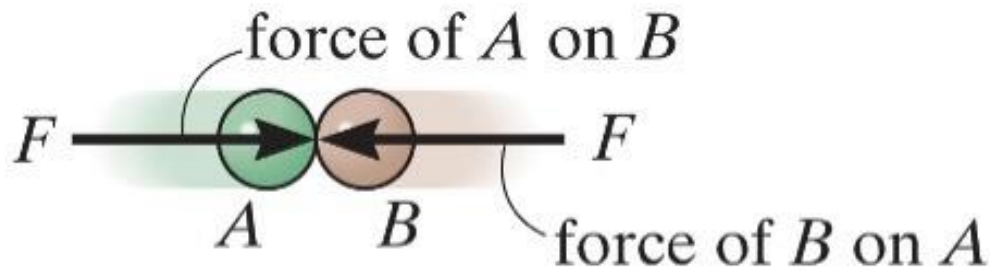
Accelerated Motion



Fundamental Concepts & Principles

- **Third Law** – The mutual forces of action and reaction between two particles are equal, opposite, and collinear.

$$\mathbf{F}_{AB} = -\mathbf{F}_{BA}$$



Action – Reaction

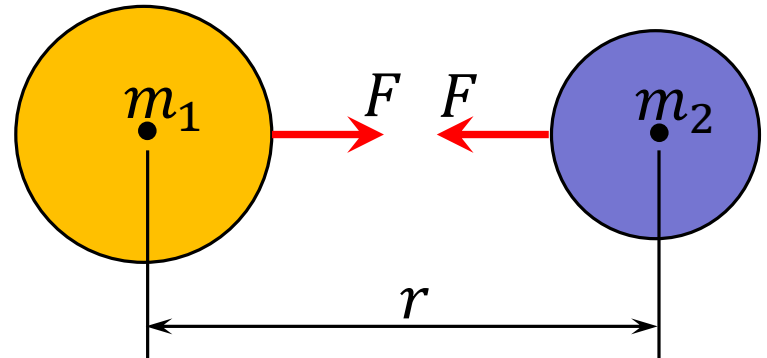


Fundamental Concepts & Principles

D. Newton's Law of Gravitational Attraction

- The gravitational attraction between any two particles (equal and opposite):

$$F = \frac{Gm_1m_2}{r^2}$$



F = force of gravitation between the two particles

G = universal gravitation constant ($6.673 \times 10^{-11} \text{ m}^3/\text{kgs}^2$)

m_1, m_2 = mass of each of the two particles

r = distance between the two particles

Fundamental Concepts & Principles

E. Weight

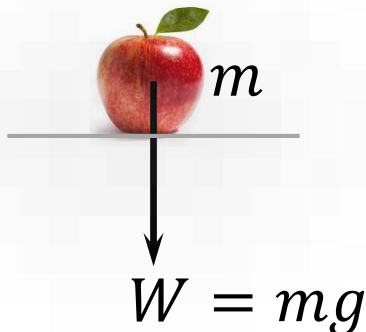
- Weight is the force on an object due to gravity.
- g is the acceleration due to gravity (gravitational intensity)

$$W = \frac{GmM_e}{r^2} \Rightarrow \boxed{W = mg}$$

$$\text{where } g = \frac{GM_e}{r^2}$$

M_e = mass of the earth

m = mass of a particle



Problems

$$G = 6.673 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$$

$$M_e = 5.972 \times 10^{24} \text{ kg}$$

$$g = 9.81 \text{ m/s}^2$$

Problem 1: Determine the force of attraction between two particles with masses of 10 kg and 18 kg, which are 0.4 m apart.

Problem 2: A satellite with a mass of 1000 kg is in orbit around the Earth, which has a mass of 6.0×10^{24} kg. The gravitational force exerted by the Earth on the satellite is 0.050 N. Find the distance from the center of the Earth to the satellite.

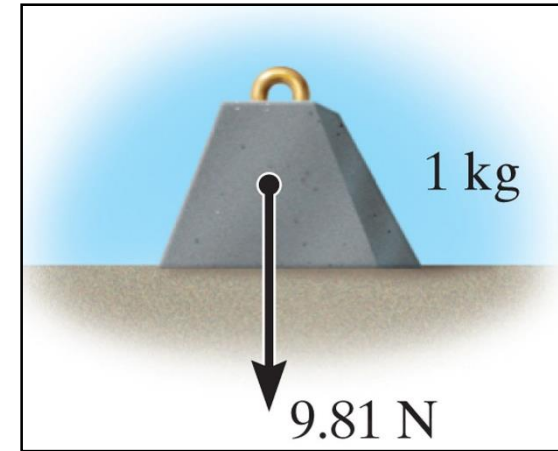
Systems of Units

- Units are arbitrary names we give to the physical quantities.
- The units of Length, Time, and Mass are called the **base units**, they are defined.
- The unit of Force is then derived, **derived unit**.

Systems of Units

International System (SI) or Metric Units

- Length – **meters (m)**, centimeters (cm), etc.
- Time – **seconds (s)**, hours (h), etc.
- Mass – **kilograms (kg)**, grams (g), etc.
- $g = 9.81 \text{ ms}^{-2}$ or N/kg
- Force is derived unit – Newton (N). $1 \text{ N} = 1 \text{ kgms}^{-2}$



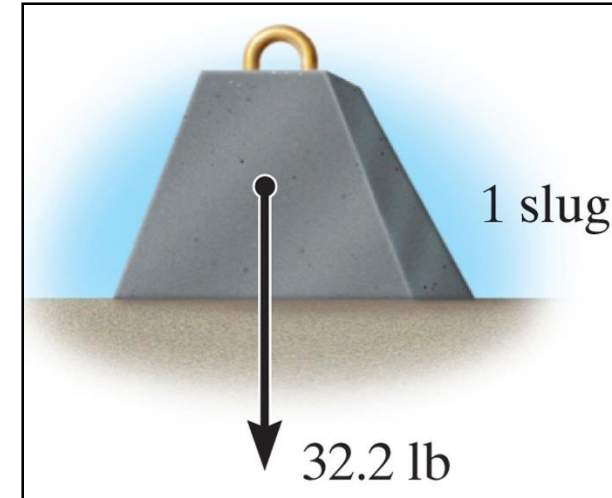
❖ Prefix:

- $1,000,000,000 = 10^9 = 1 \text{ giga} = 1 \text{ G}$
- $1,000,000 = 10^6 = 1 \text{ mega} = 1 \text{ M}$
- $1,000 = 10^3 = 1 \text{ kilo} = 1 \text{ k}$
- $0.001 = 10^{-3} = 1 \text{ milli} = 1 \text{ m}$
- $0.000\ 001 = 10^{-6} = 1 \text{ micro} = 1 \mu$
- $0.000\ 000\ 001 = 10^{-9} = 1 \text{ nano} = 1 \text{ n}$

Systems of Units

U.S. Customary (USCS) or English (FPS) Units

- Length – **feet (ft)**, inches (in), etc.
- Time – **seconds (s)**, hours (h), etc.
- Force – pounds (lb). 1000 lb = 1 kip
- $g = 32.2 \text{ ft/s}^2$
- Mass is a derived unit – **slug**



Conversion of Units

- 12 in (inches) = 1 ft
- 5280 ft = 1 mi (mile)
- 2000 lb = 1 ton
- Force: 1 lb = 4.448 N
- Mass: 1 slug = 14.59 kg
- Length: 1 ft = 0.3048 m, etc.

Numerical Calculations

Dimensional Homogeneity

- The terms of any equation must have dimensional “homogeneity”.

Rounding off Numbers and Calculations

- Use an appropriate number of significant figures (3 for answer, and 4 for intermediate calculations).

Problems

Problem 2

Experimental data yields the value $G = 6.673 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$ for the Universal Gravitational Constant. What is the force of attraction between two identical steel spheres each of mass 1 kg and whose centres are 200 mm apart?

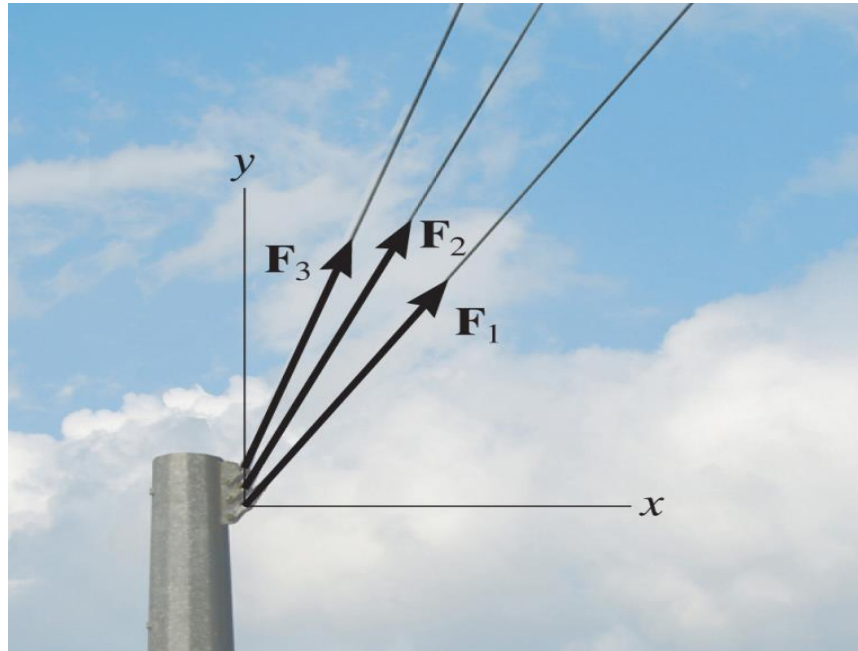
Problems

Problem 3

- (a) Convert 0.28 kg to g. (b) Convert 8 cm^2 to m^2 .
(c) Convert 20 m^3 to mm^3 .
(d) Convert 10 hours to seconds.

Topic 2

Force Vectors



Application

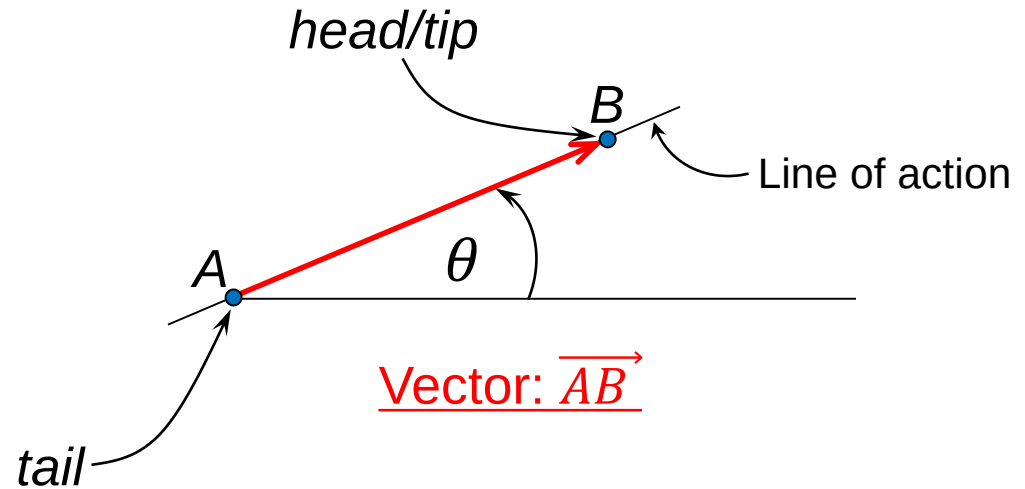
- We need to decide, for example, if the hook in Figure (b) below will fail (bend or break).
- To do this, we need to know the resultant or total force acting on the hook/rope.



Vectors Representation

- A vector is any physical quantity that requires both a *magnitude* and a *direction* for its complete description.
- Examples: force, displacement, acceleration, moment, etc.

- $\overrightarrow{BA} = -\overrightarrow{AB}$
- Vector length = magnitude
- Arrow – sense of direction
- θ defines direction



■ Notations

- Magnitude: $|\mathbf{F}|$ or F , $|\overrightarrow{AB}|$, etc.
- Vector: Printed (boldface) - $\mathbf{F}$, $\mathbf{u}$, etc.
- Handwritten work: $\overline{\mathbf{F}}$, $\underline{\mathbf{u}}$, etc.

Vector Addition (Graphical)

■ Operations

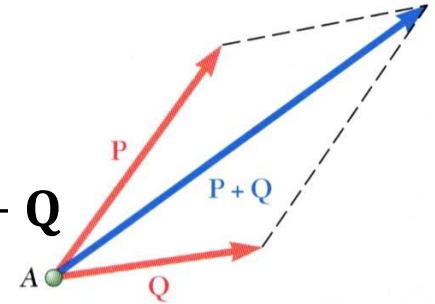
○ Addition and Subtraction

- Negative vectors
- Parallelogram law of addition
- Triangle rule
- Resultant (vector)
- Trapezoid rule

○ Parallel vectors

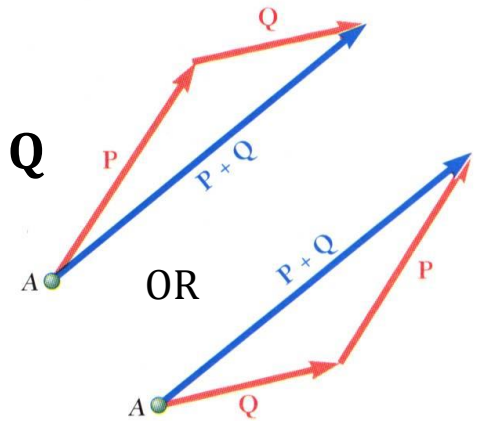
○ Equal vectors

$$\mathbf{F}_R = \mathbf{P} + \mathbf{Q}$$



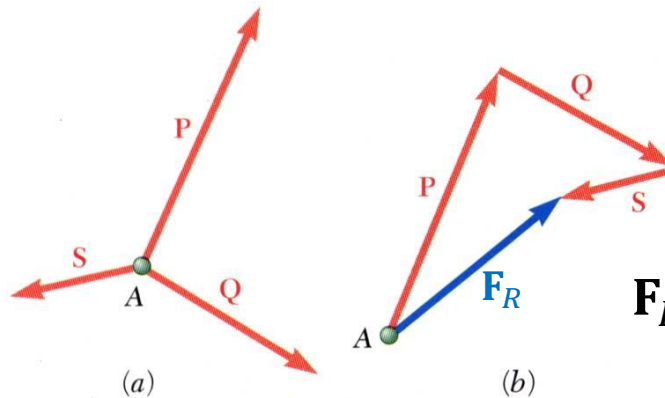
Parallelogram law

$$\mathbf{F}_R = \mathbf{P} + \mathbf{Q}$$



OR

Triangle rule



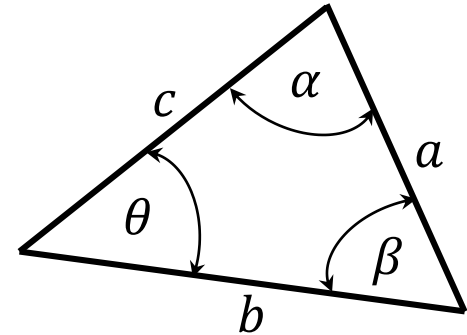
$$\mathbf{F}_R = \mathbf{P} + \mathbf{Q} + \mathbf{S}$$

Trapezoid rule

Cosine and Sine Laws

- Cosine law

$$a^2 = b^2 + c^2 - 2(b)(c)\cos\theta$$



- Sine law

$$\frac{a}{\sin\theta} = \frac{b}{\sin\alpha} = \frac{c}{\sin\beta}$$

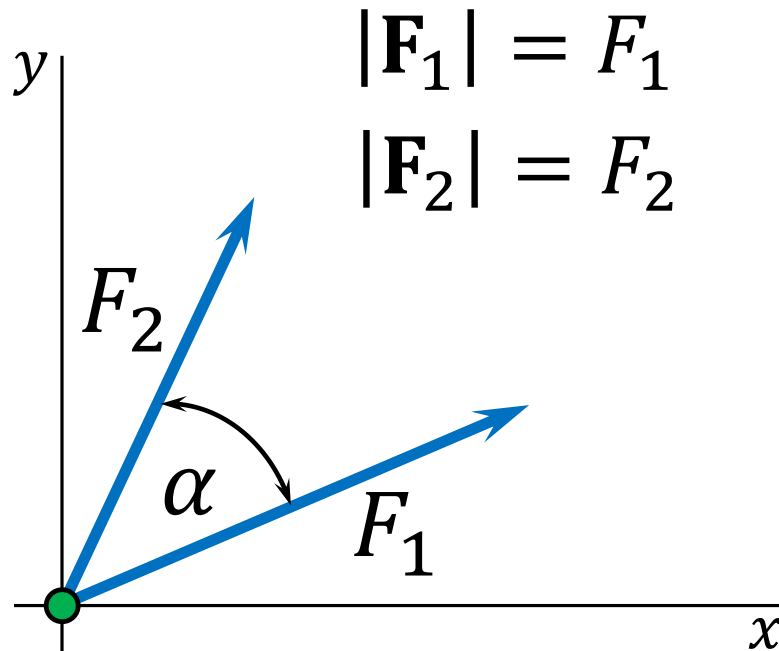
OR

$$\frac{\sin\theta}{a} = \frac{\sin\alpha}{b} = \frac{\sin\beta}{c}$$

Parallelogram and Triangle Rule

Application of Parallelogram Law and Triangle Rule

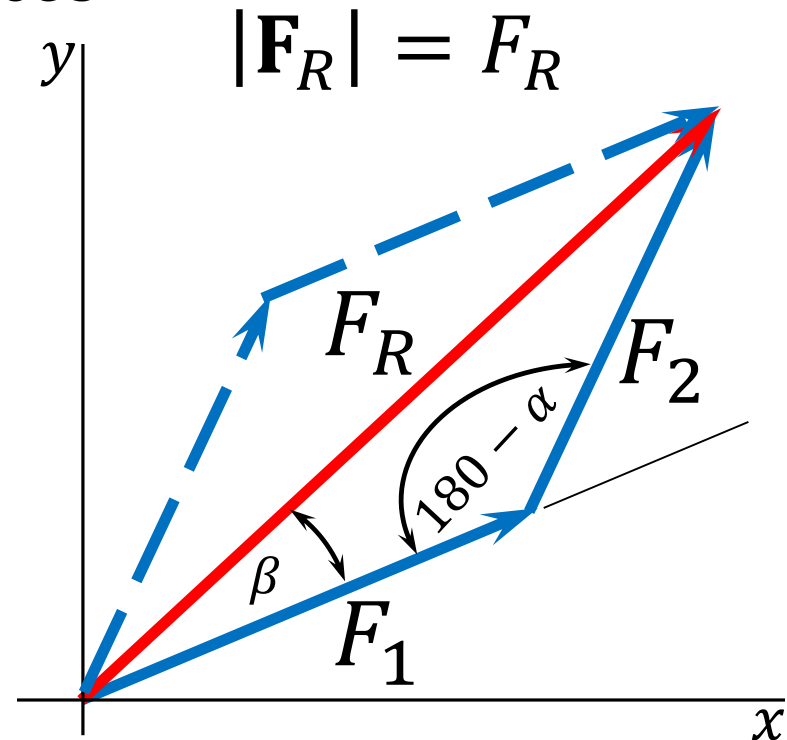
- To determine resultant of two forces



Magnitude

$$F_R^2 = F_1^2 + F_2^2 - 2F_1F_2\cos(180 - \alpha)$$

$$F_R = ?$$



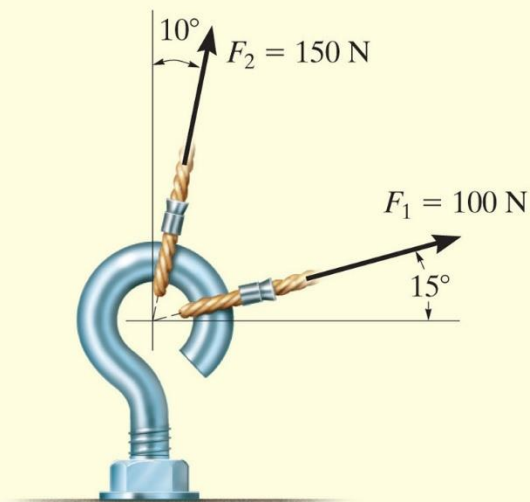
Direction

$$\frac{\sin\beta}{F_2} = \frac{\sin(180 - \alpha)}{F_R} \Rightarrow \beta = ?$$

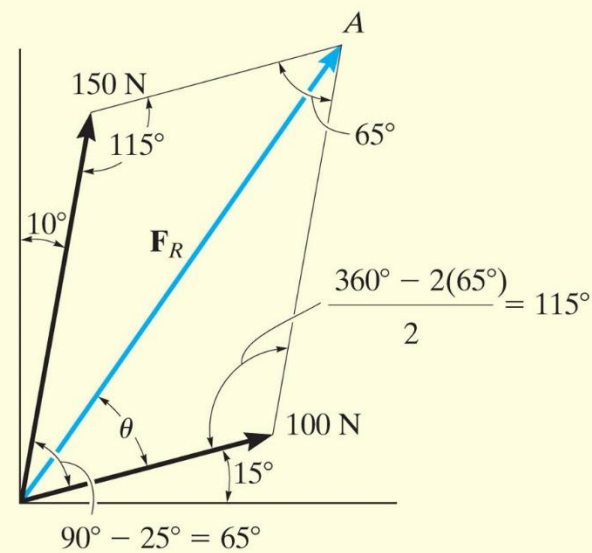
Direction, $\theta = ?$

EXAMPLE 2.1

The screw eye in Fig. 2–11a is subjected to two forces, $\mathbf{F}_1$ and $\mathbf{F}_2$. Determine the magnitude and direction of the resultant force.



(a)



(b)

SOLUTION

Parallelogram Law. The parallelogram is formed by drawing a line from the head of $\mathbf{F}_1$ that is parallel to $\mathbf{F}_2$, and another line from the head of $\mathbf{F}_2$ that is parallel to $\mathbf{F}_1$. The resultant force $\mathbf{F}_R$ extends to where these lines intersect at point A, Fig. 2–11b. The two unknowns are the magnitude of $\mathbf{F}_R$ and the angle θ (theta).

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EXAMPLE 2.1 CONTINUED

Trigonometry. From the parallelogram, the vector triangle is constructed, Fig. 2–11c. Using the law of cosines

$$\begin{aligned} F_R &= \sqrt{(100 \text{ N})^2 + (150 \text{ N})^2 - 2(100 \text{ N})(150 \text{ N}) \cos 115^\circ} \\ &= \sqrt{10\,000 + 22\,500 - 30\,000(-0.4226)} = 212.6 \text{ N} \\ &= 213 \text{ N} \end{aligned}$$

Ans.

Applying the law of sines to determine θ ,

$$\begin{aligned} \frac{150 \text{ N}}{\sin \theta} &= \frac{212.6 \text{ N}}{\sin 115^\circ} & \sin \theta &= \frac{150 \text{ N}}{212.6 \text{ N}} (\sin 115^\circ) \\ & & \theta &= 39.8^\circ \end{aligned}$$

Thus, the direction ϕ (phi) of $\mathbf{F}_R$, measured from the horizontal, is

$$\phi = 39.8^\circ + 15.0^\circ = 54.8^\circ$$

Ans.

NOTE: The results seem reasonable, since Fig. 2–11*b* shows $\mathbf{F}_R$ to have a magnitude larger than its components and a direction that is between them.

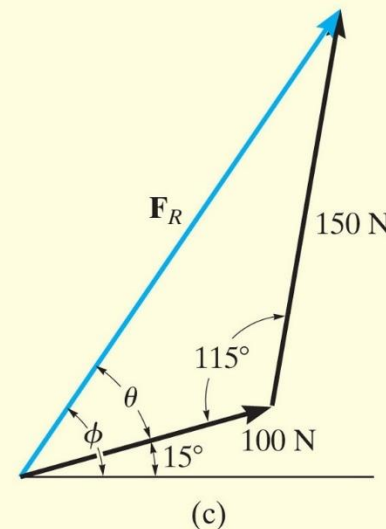


Fig. 2–11

EXAMPLE 2.2

Resolve the horizontal 600-lb force in Fig. 2–12a into components acting along the u and v axes and determine the magnitudes of these components.

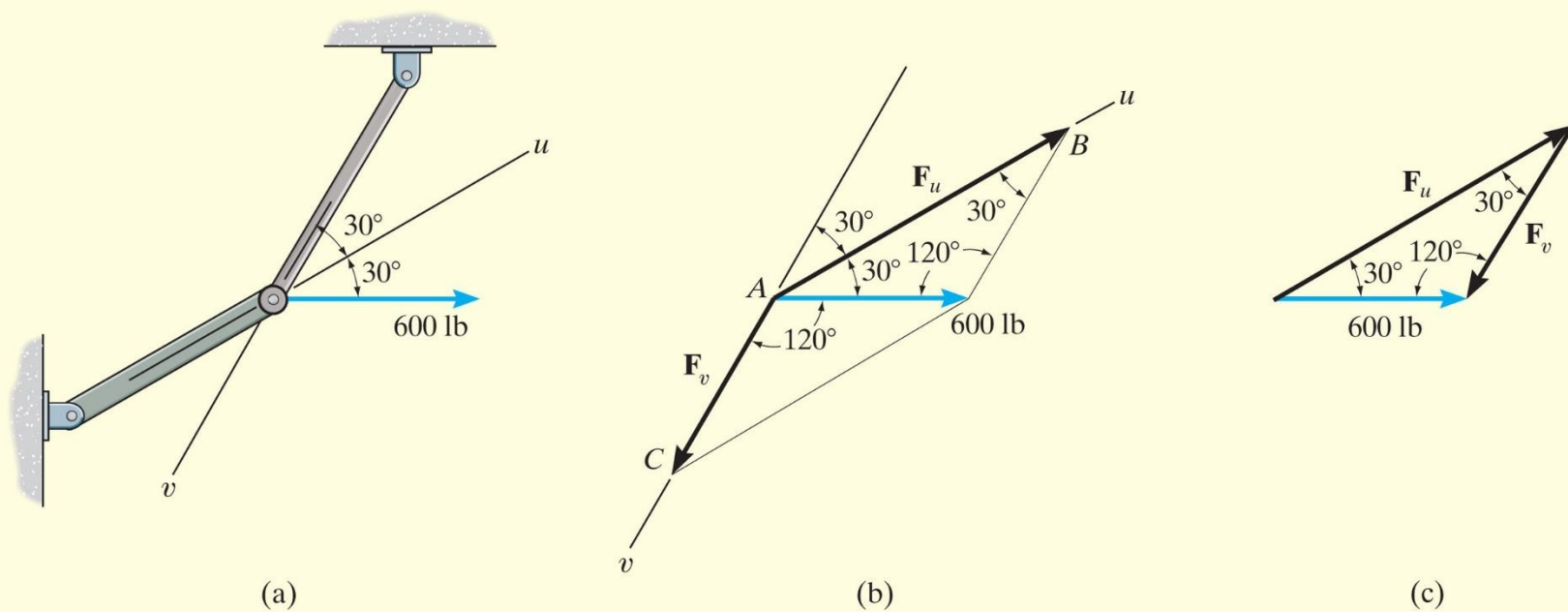


Fig. 2–12

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EXAMPLE 2.2 CONTINUED

SOLUTION

The parallelogram is constructed by extending a line from the *head* of the 600-lb force parallel to the v axis until it intersects the u axis at point B , Fig. 2–12*b*. The arrow from A to B represents $\mathbf{F}_u$. Similarly, the line extended from the head of the 600-lb force drawn parallel to the u axis intersects the v axis at point C , which gives $\mathbf{F}_v$.

The vector addition using the triangle rule is shown in Fig. 2–12*c*. The two unknowns are the magnitudes of $\mathbf{F}_u$ and $\mathbf{F}_v$. Applying the law of sines,

$$\frac{F_u}{\sin 120^\circ} = \frac{600 \text{ lb}}{\sin 30^\circ}$$
$$F_u = 1039 \text{ lb} \quad \text{Ans.}$$

$$\frac{F_v}{\sin 30^\circ} = \frac{600 \text{ lb}}{\sin 30^\circ}$$
$$F_v = 600 \text{ lb} \quad \text{Ans.}$$

NOTE: The result for F_u shows that sometimes a component can have a greater magnitude than the resultant.

EXAMPLE 2.3

Determine the magnitude of the component force $\mathbf{F}$ in Fig. 2–13*a* and the magnitude of the resultant force $\mathbf{F}_R$ if $\mathbf{F}_R$ is directed along the positive y axis.

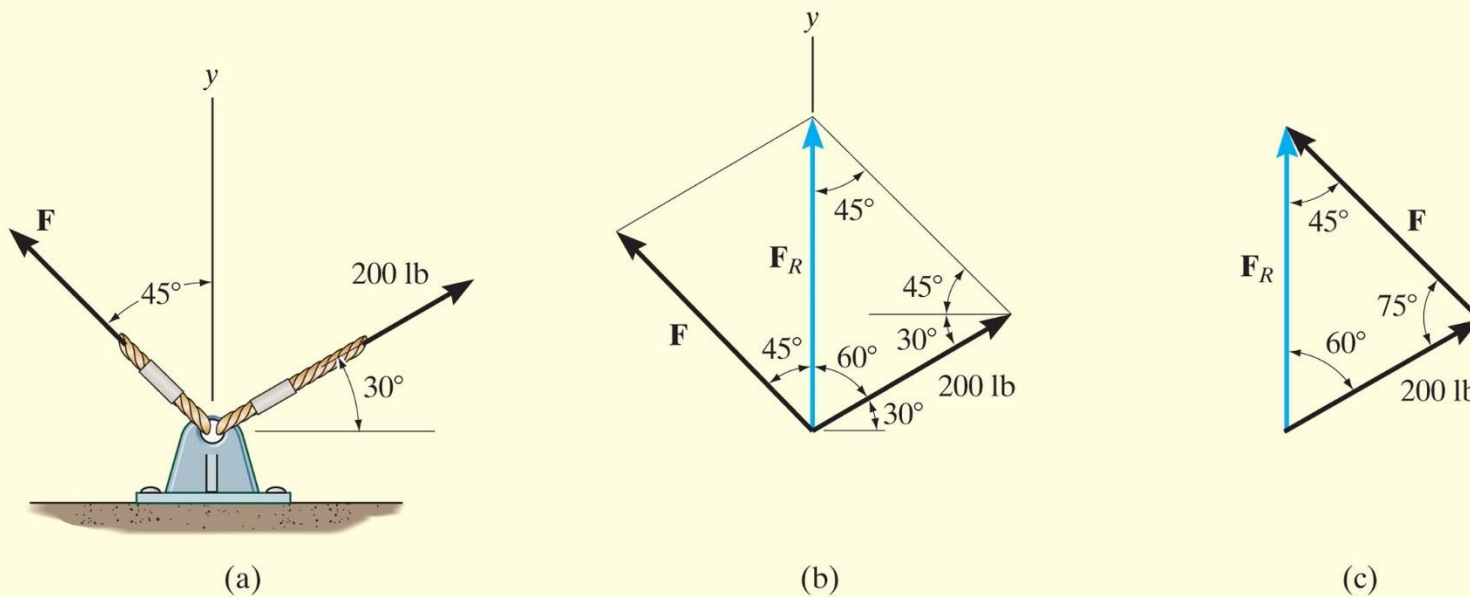


Fig. 2–13

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EXAMPLE**2.3 CONTINUED****SOLUTION**

The parallelogram law of addition is shown in Fig. 2–13*b*, and the triangle rule is shown in Fig. 2–13*c*. The magnitudes of $\mathbf{F}_R$ and $\mathbf{F}$ are the two unknowns. They can be determined by applying the law of sines.

$$\frac{F}{\sin 60^\circ} = \frac{200 \text{ lb}}{\sin 45^\circ}$$

$$F = 245 \text{ lb}$$

Ans.

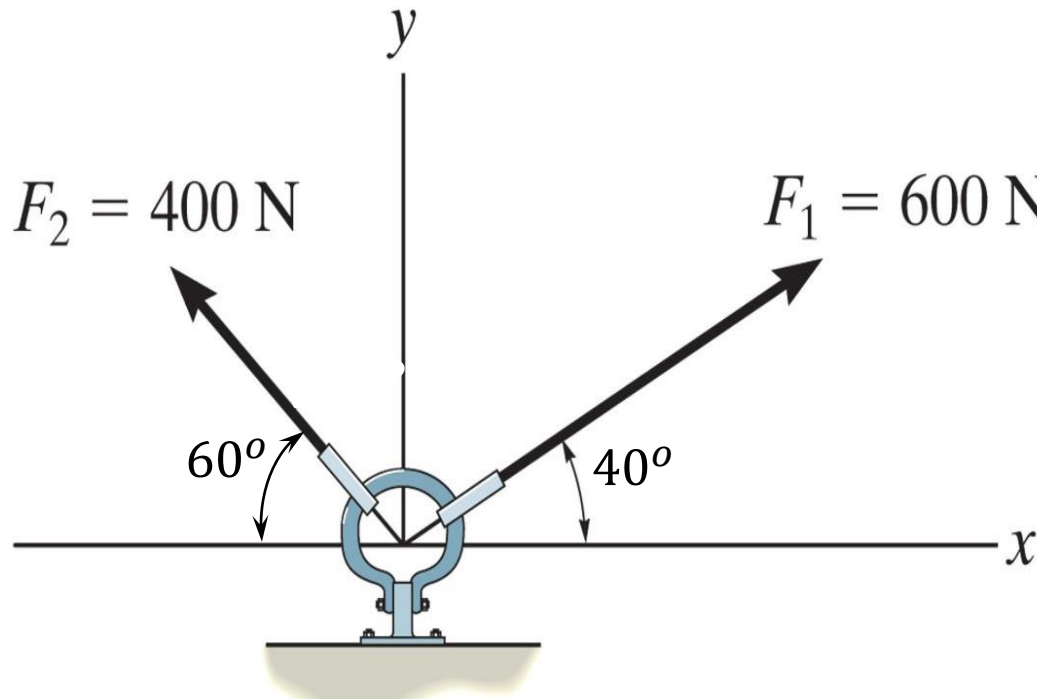
$$\frac{F_R}{\sin 75^\circ} = \frac{200 \text{ lb}}{\sin 45^\circ}$$

$$F_R = 273 \text{ lb}$$

Ans.

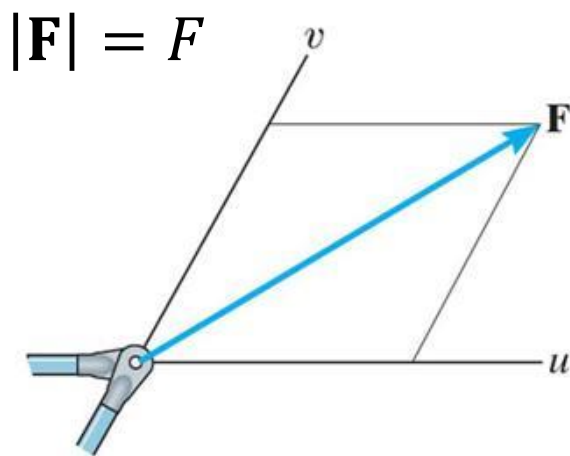
Example 1

- Application of Parallelogram Law and Triangle Rule
 - Determine the magnitude of the resultant force and its direction, measured counter-clockwise from the positive x -axis.

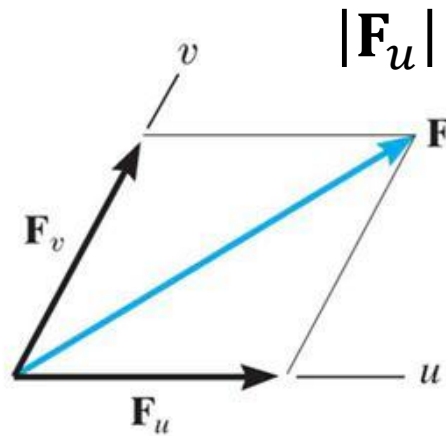


Components of a Force along Arbitrary Axes

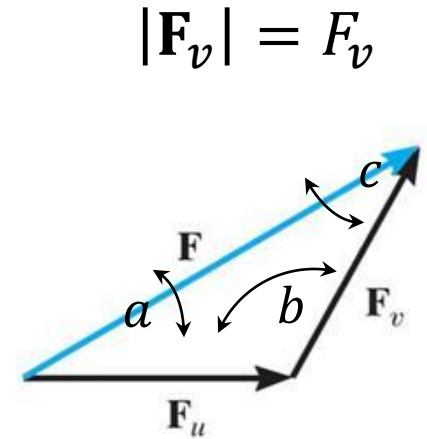
- Application of Parallelogram Law and Triangle Rule
 - To resolve a force into components acting along two axes
 - It is kind of like using the parallelogram law in reverse.



(a)



(b)



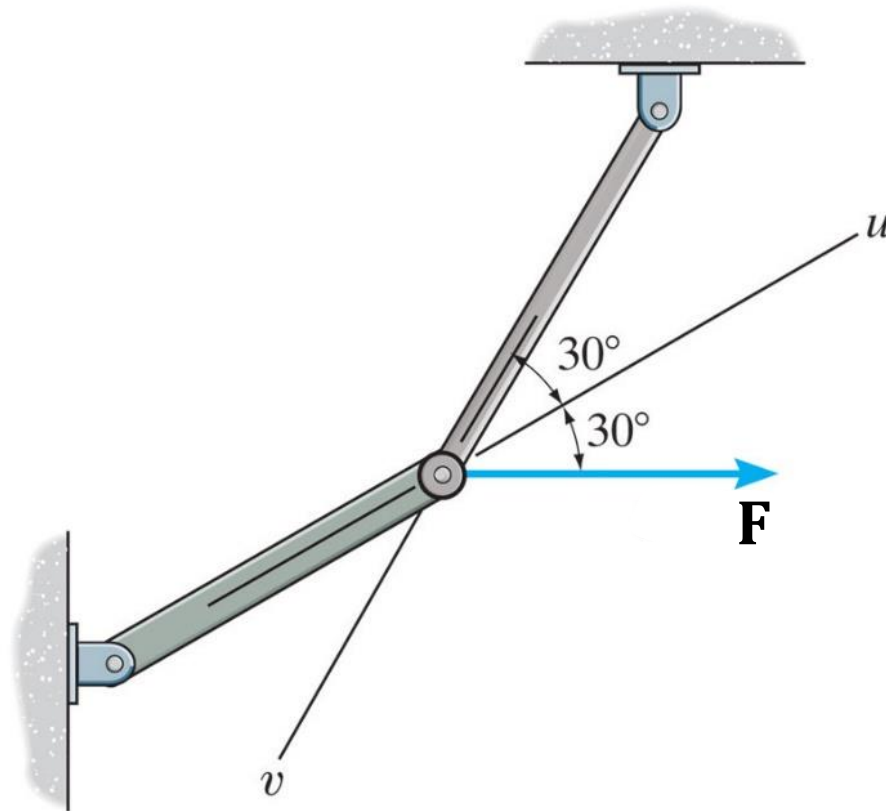
(c)

$$\frac{F_u}{\sin c} = \frac{F}{\sin b} \Rightarrow F_u = ?$$

$$\frac{F_v}{\sin a} = \frac{F}{\sin b} \Rightarrow F_v = ?$$

Example 2

- Application of Parallelogram Law and Triangle Rule
 - If magnitude of the force $\mathbf{F}$ is 200 N, resolve this force into components acting along the u and v axes.



Rectangular Components of 2D Forces

■ Resolving a Force into Rectangular Components

- Components of a force along x -axis (horizontal) and y -axis (vertical)

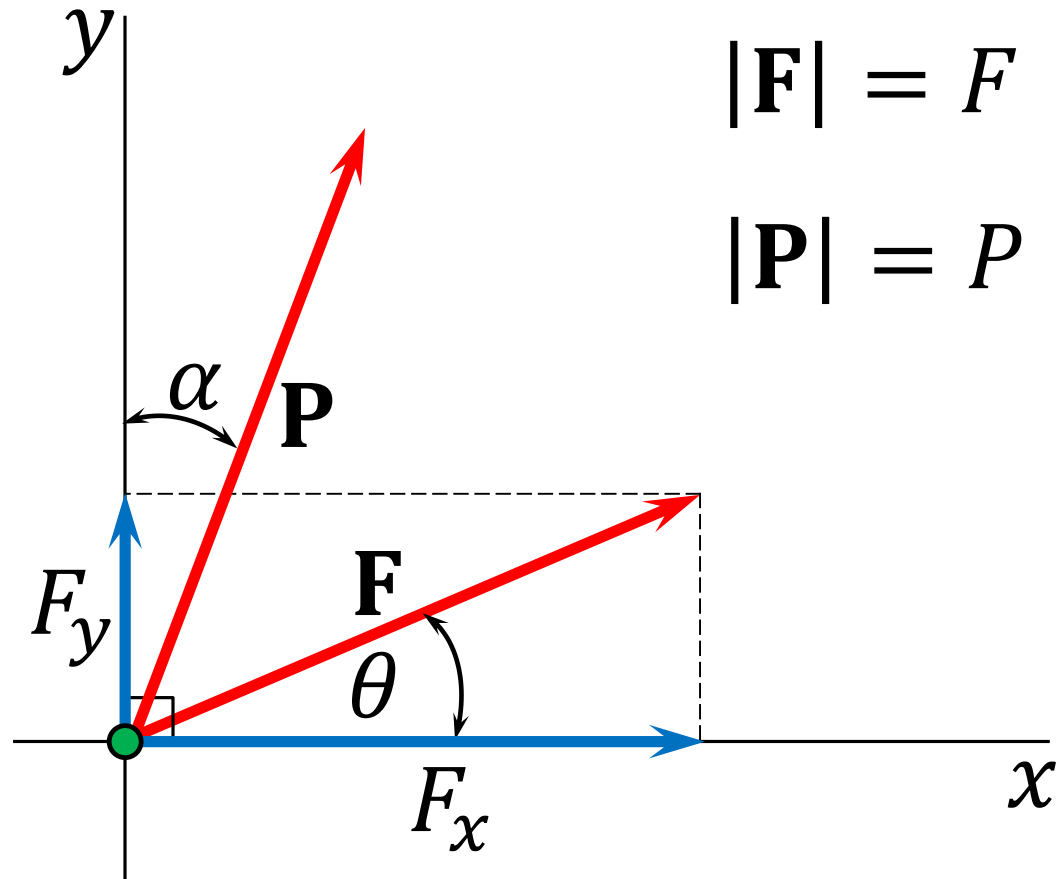
- Scalar Notation

$$F_x = F \cos \theta$$

$$F_y = F \sin \theta$$

$$P_x = P \sin \alpha$$

$$P_y = P \cos \alpha$$



Note: If a component points in the negative direction of the x and y axes, then it has a negative value.

Rectangular Components of 2D Forces

■ Resolving a Force into Rectangular Components

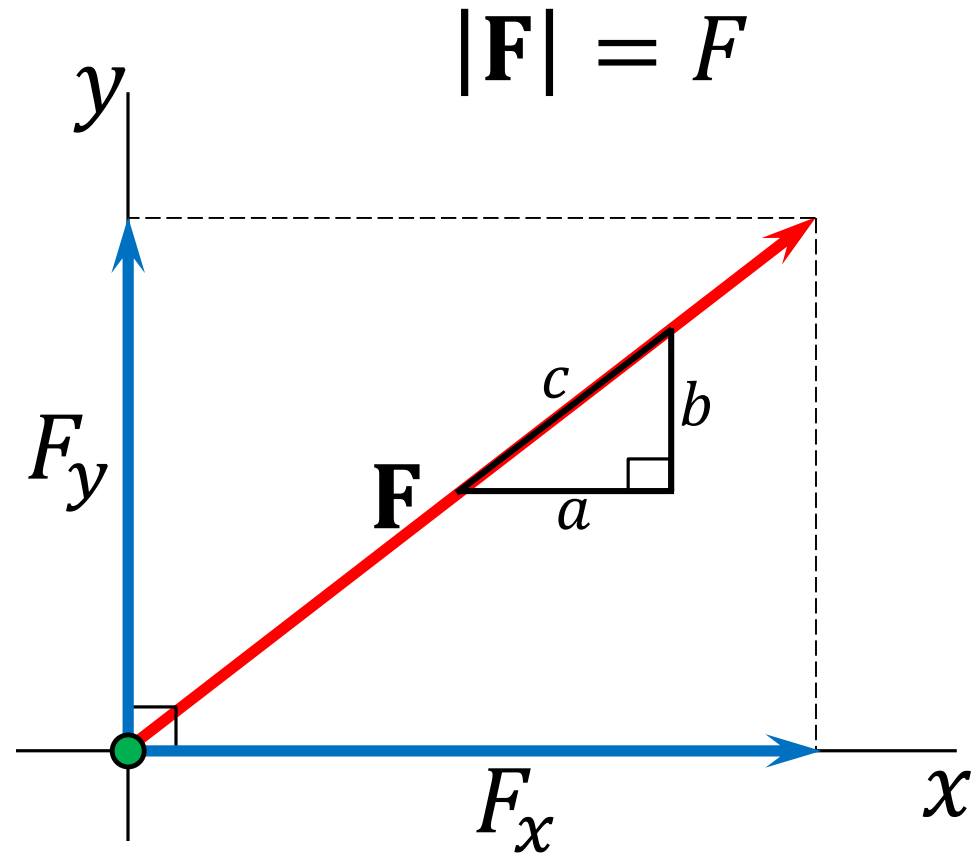
- Components of a force along x -axis (horizontal) and y -axis (vertical)

○ Scalar Notation

- Using “Slope triangle”

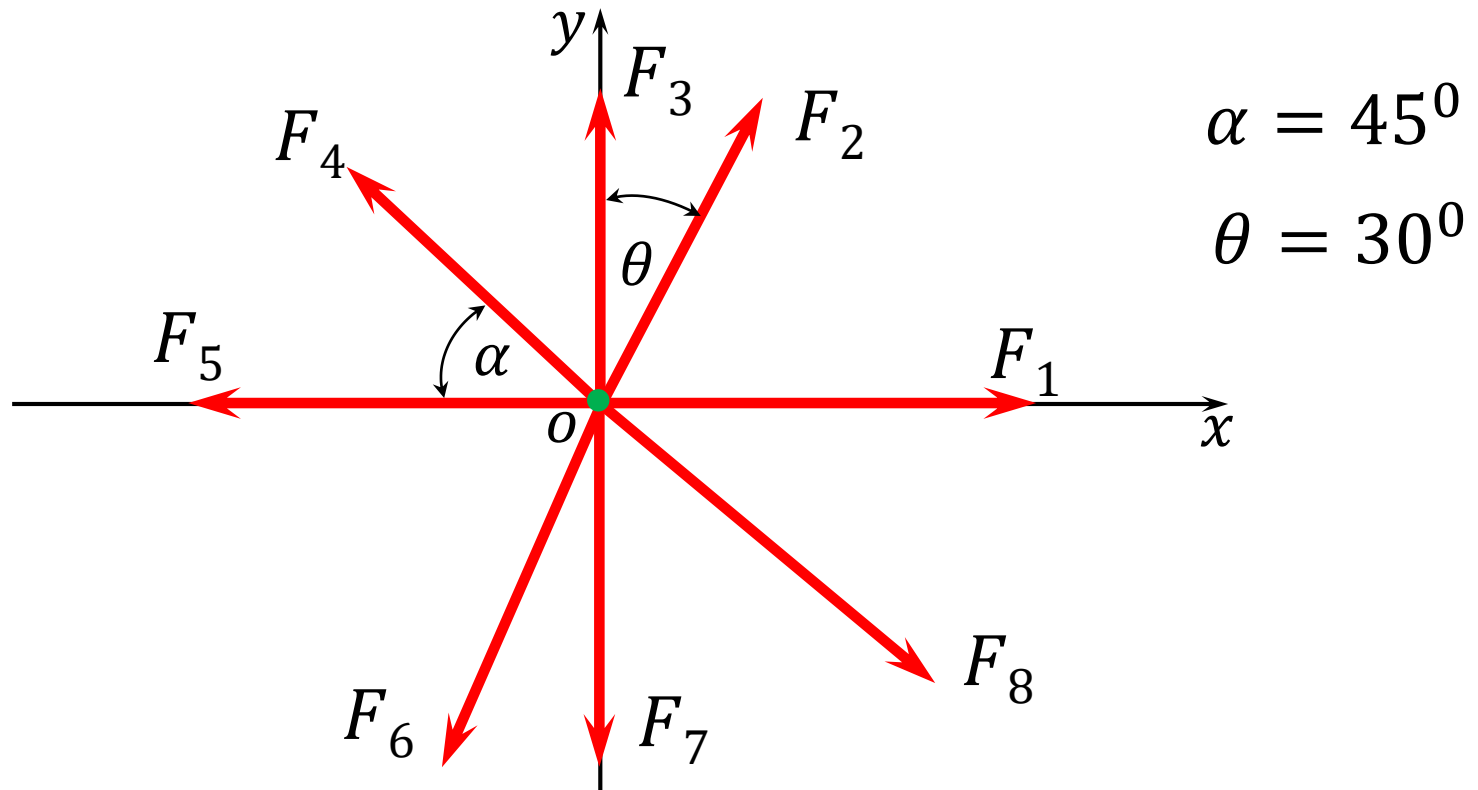
$$F_x = F \left(\frac{a}{c} \right)$$

$$F_y = F \left(\frac{b}{c} \right)$$



Example 3

- Resolving a Force into Rectangular Components
 - Determine the x and y components of each force $\mathbf{F}$ in the diagram if its magnitude is 10 N. *Give the exact values for the answers.*



Vector Notation of a Force

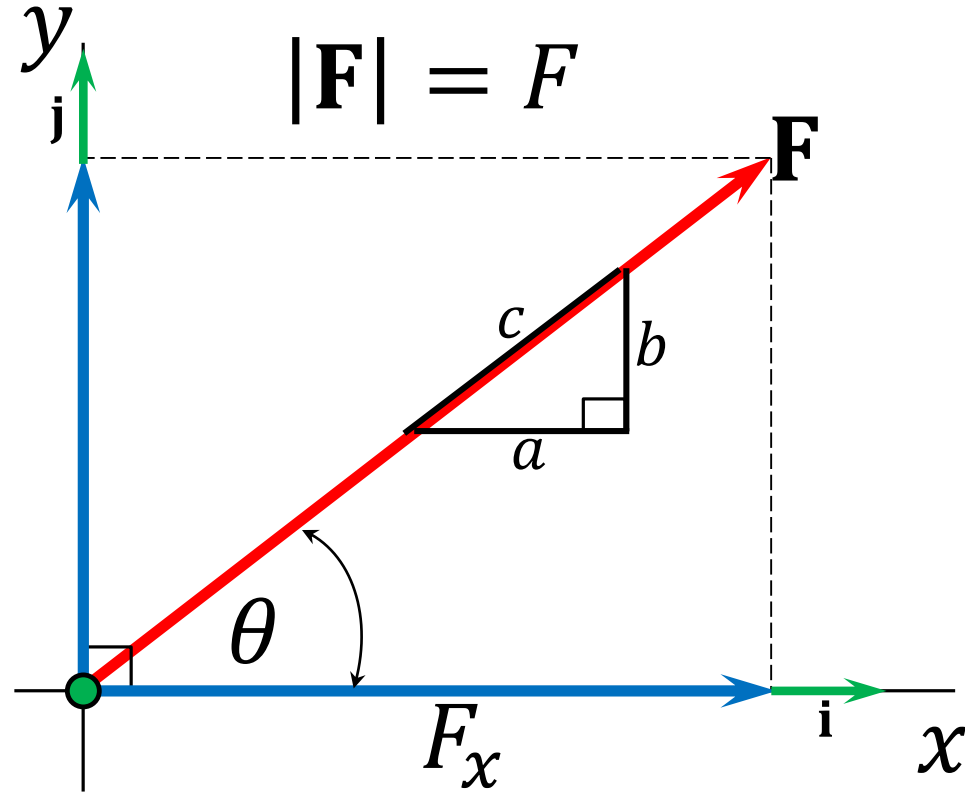
- Cartesian Vector Notation

- Column vector format

$$F_x = F \cos \theta$$

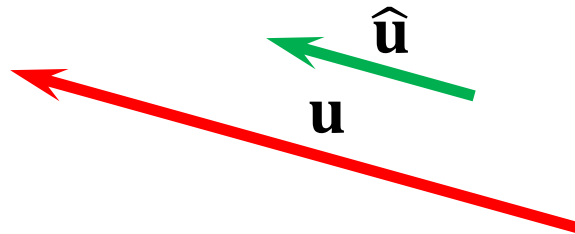
$$F_y = F \sin \theta$$

$$\mathbf{F} = \begin{pmatrix} F \cos \theta \\ F \sin \theta \end{pmatrix}$$



- Unit vector format

$$\hat{\mathbf{u}} = \frac{\mathbf{u}}{|\mathbf{u}|}, |\hat{\mathbf{u}}| = 1$$



Vector Notation of a Force

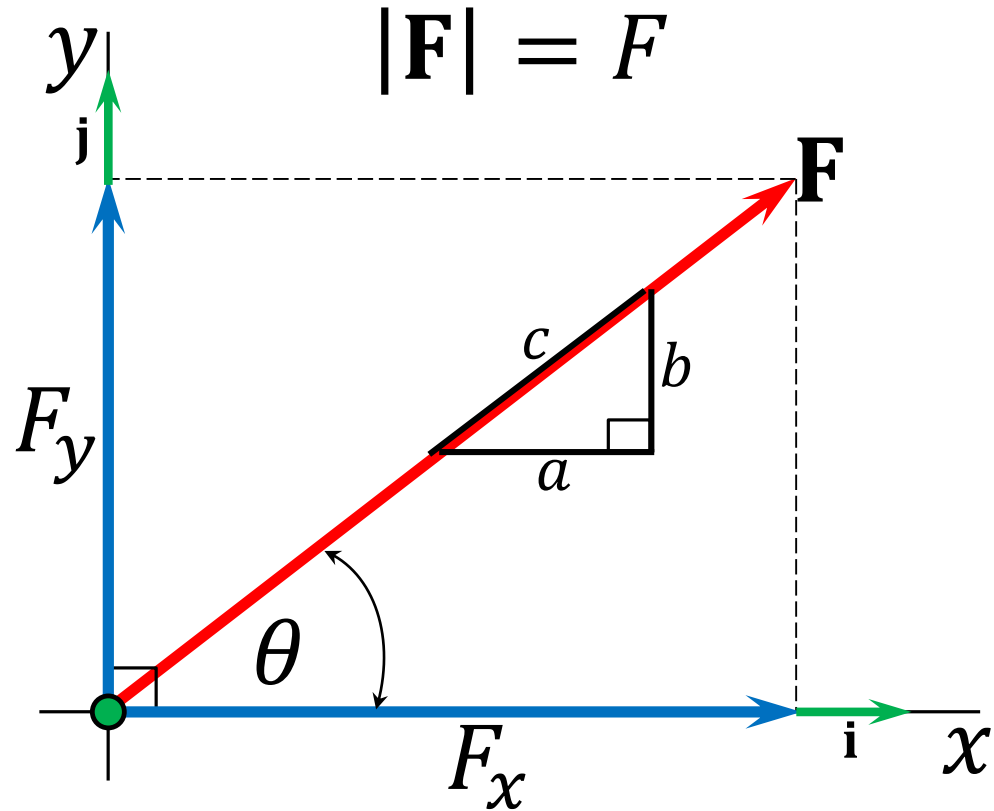
■ Unit Vector Notation

- 'Special' unit vectors

x -direction: $\hat{\mathbf{i}} = \mathbf{i}$

y -direction: $\hat{\mathbf{j}} = \mathbf{j}$

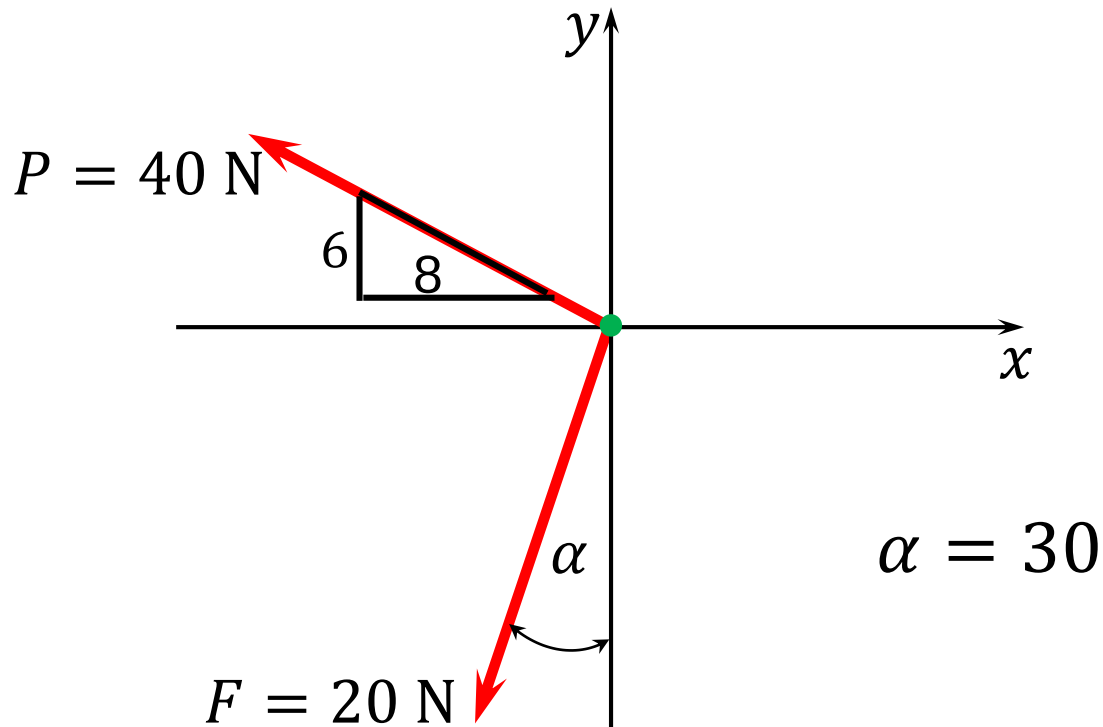
$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j}$$



Example 4

■ Cartesian Vector Notation

- Determine the forces **P** and **F** in Cartesian vector notation form. *Give the exact values for the answers.*



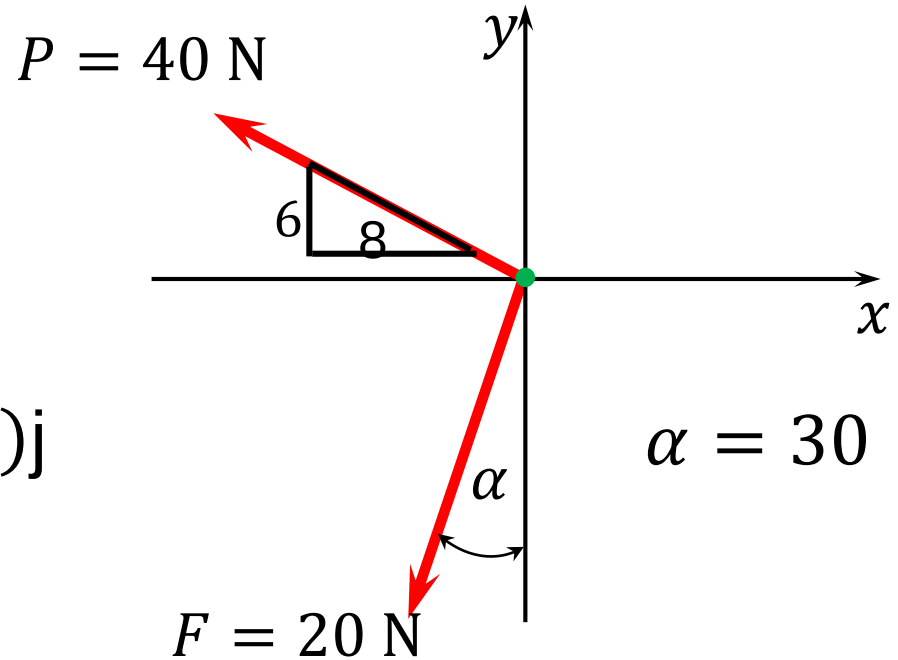
Example 4 - Solution

$$\mathbf{F} = F_x \mathbf{i} + F_y \mathbf{j}$$

$$\mathbf{F} = (-20 \sin \alpha) \mathbf{i} + (-20 \cos \alpha) \mathbf{j}$$

$$\mathbf{F} = (-20 \sin 30^\circ) \mathbf{i} + (-20 \cos 30^\circ) \mathbf{j}$$

$$\mathbf{F} = (-10 \mathbf{i} - 17.321 \mathbf{j}) \text{ N}$$

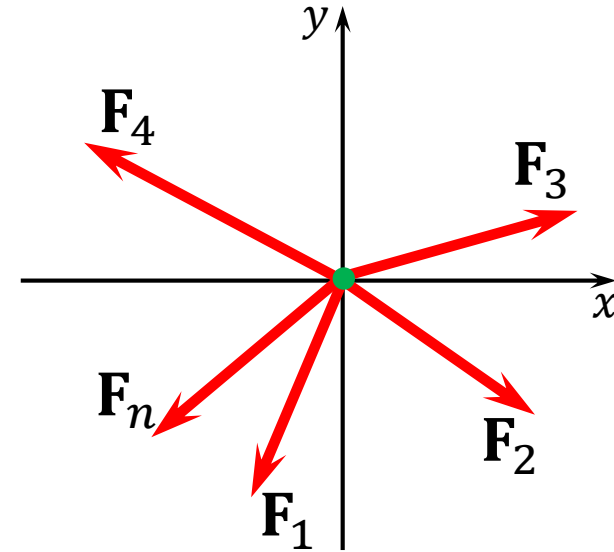


Addition of Concurrent Coplanar Forces

○ Method 1: Scalar Notation

$$\left(\begin{smallmatrix} + \\ \rightarrow \end{smallmatrix}\right) (F_R)_x = \sum F_x = \pm F_{1x} \pm F_{2x} \pm \cdots \pm F_{nx}$$

$$(+\uparrow) (F_R)_y = \sum F_y = \pm F_{1y} \pm F_{2y} \pm \cdots \pm F_{ny}$$



○ Method 2: Cartesian Vector Notation

$$\begin{aligned} \mathbf{F}_R &= \sum \mathbf{F} = (\pm F_{1x} \mathbf{i} \pm F_{1y} \mathbf{j}) + (\pm F_{2x} \mathbf{i} \pm F_{2y} \mathbf{j}) + \cdots + (\pm F_{nx} \mathbf{i} \pm F_{ny} \mathbf{j}) \\ &= \pm \sum F_x \mathbf{i} \pm \sum F_y \mathbf{j} = \pm (F_R)_x \mathbf{i} \pm (F_R)_y \mathbf{j} \end{aligned}$$

Addition of Concurrent Coplanar Forces

■ Resultant of Several Concurrent Coplanar Forces

○ Magnitude

$$F_R = \sqrt{((F_R)_x)^2 + ((F_R)_y)^2}$$

○ Direction

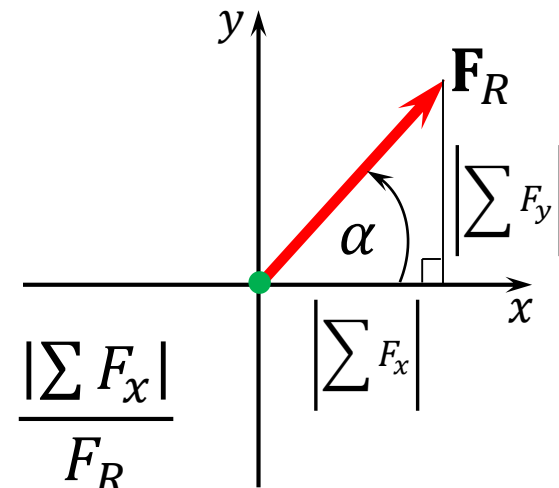
- $\mathbf{F}_R$ has to be drawn first when determining the direction, then find α . The direction α can then be found.

$$\tan \alpha = \frac{|\sum F_y|}{|\sum F_x|}$$

$$\sin \alpha = \frac{|\sum F_y|}{F_R}$$

$$\cos \alpha = \frac{|\sum F_x|}{F_R}$$

$\therefore$ the direction, $\alpha = ?$

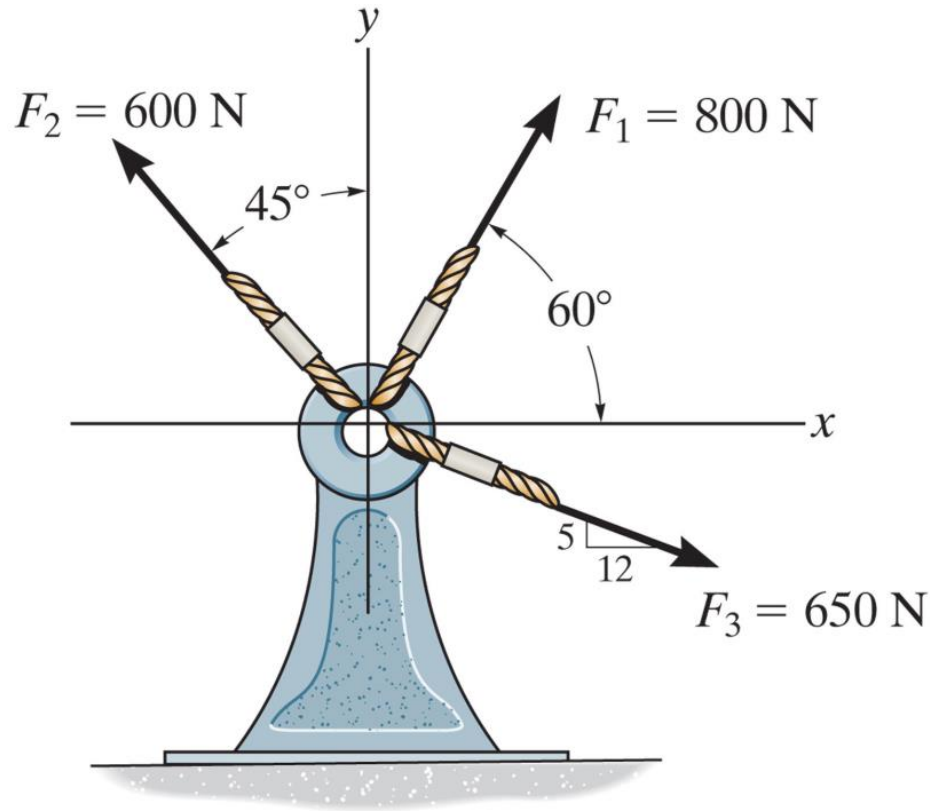


Note: The use of the scalar notation is more efficient since the components can be found directly. Also, if the forces are only two, then in addition to the above methods, the parallelogram law can also be applied.

Example 5

■ Resultant of Several Concurrent Coplanar Forces –Scalar Method

Determine the magnitude and direction of the resultant force of the three concurrent forces acting on the bracket.



Example 5 - Solution

$$\left(\begin{smallmatrix} + \\ \rightarrow \end{smallmatrix}\right) (F_R)_x = \sum F_x$$

$$(F_R)_x = 800\cos 60^\circ - 600\sin 45^\circ + 650\left(\frac{12}{13}\right)$$

$$(F_R)_x = 575.736 \text{ N}$$

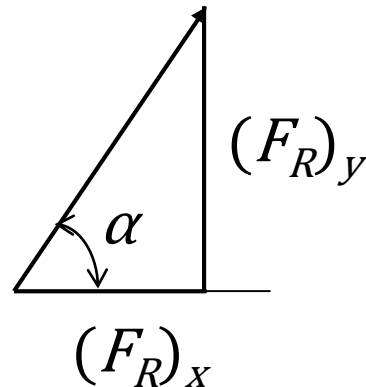
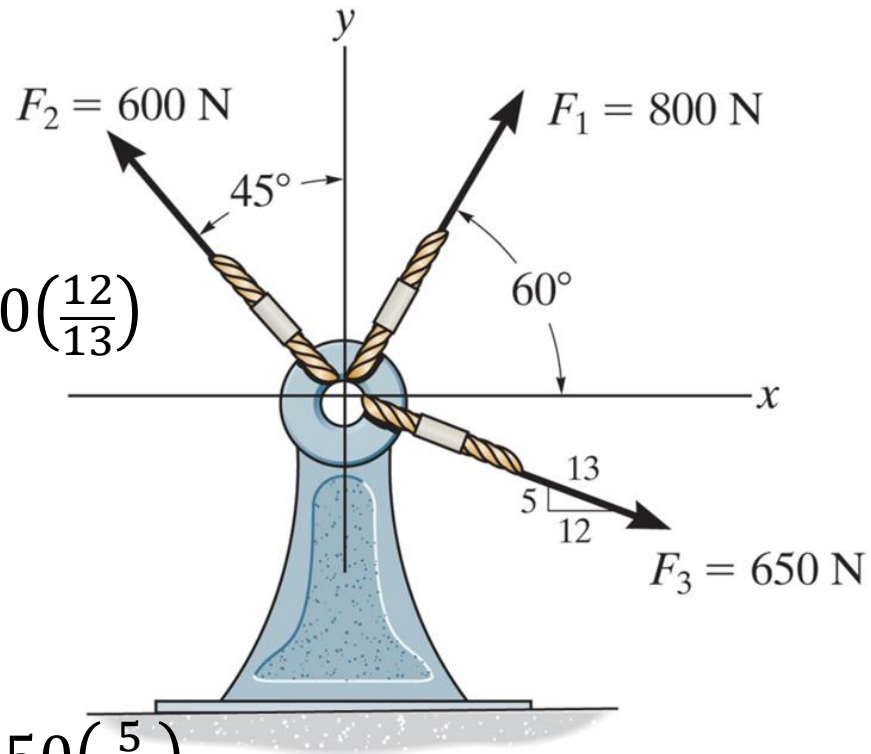
$$(+\uparrow) (F_R)_y = \sum F_y$$

$$(F_R)_y = 800\sin 60^\circ + 600\cos 45^\circ - 650\left(\frac{5}{13}\right)$$

$$(F_R)_y = 846.251 \text{ N}$$

$$F_R = \sqrt{575.736^2 + 846.251^2}$$

$$F_R = 1023.53 \text{ N}$$



$$\tan \alpha = \frac{846.251}{575.736}$$

$$\alpha = 55.8^\circ$$

Example 6

- Resultant of Several Concurrent Coplanar Forces – Vector Method

Given $\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})$ N, $\mathbf{F}_2 = -8\mathbf{j}$ N, $\mathbf{F}_3 = -8\mathbf{i} - 5\mathbf{j}$, and $\mathbf{F}_4 = -6\mathbf{i}$, determine the resultant of the force system.

Example 6 - Solution

Given $\mathbf{F}_1 = (2\mathbf{i} + 3\mathbf{j})$ N, $\mathbf{F}_2 = -8\mathbf{j}$ N, $\mathbf{F}_3 = -8\mathbf{i} - 5\mathbf{j}$, and $\mathbf{F}_4 = -6\mathbf{i}$, determine the resultant of the force system.

$$\mathbf{F}_R = \sum \mathbf{F} = (2\mathbf{i} + 3\mathbf{j}) + (-8\mathbf{j}) + (-8\mathbf{i} - 5\mathbf{j}) + (-6\mathbf{i})$$

$$\mathbf{F}_R = (2\mathbf{i} - 8\mathbf{i} - 6\mathbf{i}) + (3\mathbf{j} - 8\mathbf{j} - 5\mathbf{j})$$

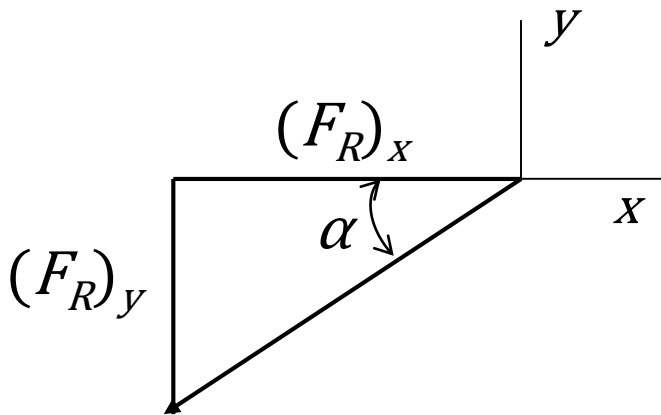
$$\mathbf{F}_R = (-12\mathbf{i} - 10\mathbf{j}) \text{ N}$$

$$\tan \alpha = \frac{|(F_R)_y|}{|(F_R)_x|}$$

$$\tan \alpha = \frac{10}{12}$$

$$\alpha = 39.8^\circ$$

$$\begin{aligned} \text{Direction of } F_R &= 270^\circ + 39.8^\circ \\ &= 309.8^\circ \end{aligned}$$



Example 7

■ Resultant of Several Concurrent Coplanar Forces

Four forces act on bolt A as shown. Determine the resultant of the force on the bolt.

